AERO40003 Computing and Numerical Methods 1
Part 2: Numerical Methods
Computing Lab 2

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Problem 2.1

In this problem we introduce the Bisection Method and implement it to find the root of a function $f(x)$. Take $f(x) = x^3 + 2x - 1$. Hint: the root lies in the region $x : (0, 1)$.

Note: As in the class handout, this function is easily solvable but the aim of the exercise is the introduction of the Bisection method rather than finding the root of a complicated function.

Bisection method

Given values $f(x1) \times f(x2) < 0$ we know that if the function is continuous, then there must be a root at c where $x1 < c < x2$.

Pseudo-Code for Bisection method

```
REPEAT
set x3 = (x1+x2)/2
     IF f(x3) × f(x1) < 0
          SET x2 = x3
          ELSE
          SET x1 = x3
     END
UNTIL |x1-x2|/2 < tolerance value
```

Implement this pseudo-code in MATLAB to find the root of $f(x)$.

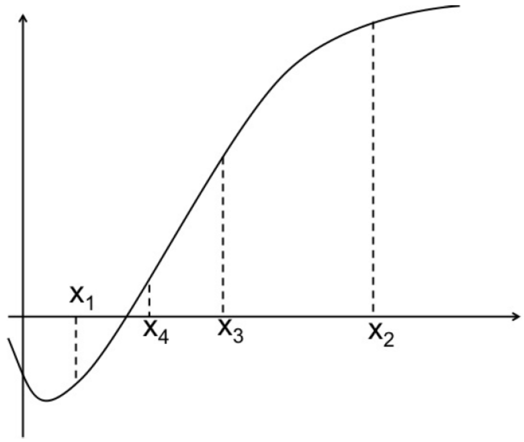

Figure 1: A schematic showing the Bisection root-finding scheme

Problem 2.2

We're again going to look at finding the root of f(x) from Problem 1, this time using Newton's method.

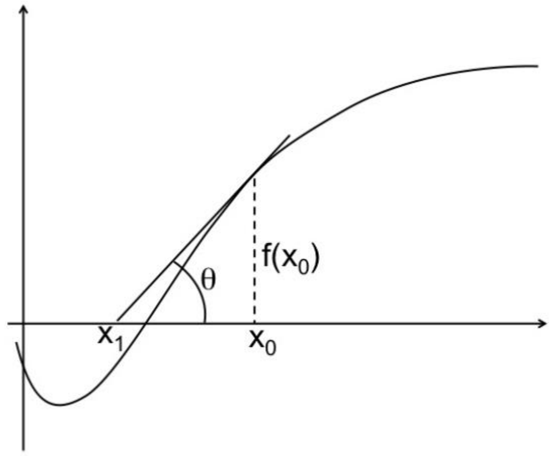

Figure 2: A schematic showing the Newton root-finding scheme

Pseudo-Code for Newton's method based on the equation in the class handout:

```
Compute f(x0) and f'(x0).
REPEAT
     SET x1 = x0 - f(x0)/f`(x0)
     SET x0 = x1 %overwrite
UNTIL |f(x0)| < tolerance value
```

Problem 2.3

Write codes in MATLAB to find the smallest positive root of the following expressions using a) the Bisection method, b) Newton's method and c) the Secant method. Use appropriate starting guesses.

$f(x) = x^4 - 2x - 1$

$f(x) = 2 - \sin 2x - e^x$

What starting guess might cause Newton's method to fail in each of the cases?

Problem 2.4

Lee and Duffy (1976) relate the friction factor for flow of a suspension of fibrous particles to the Reynolds number by this empirical equation:

$$\frac{1}{\sqrt{f}} = \left(\frac{1}{k}\right) \log_e Re\sqrt{f} + \left(14 - \frac{5.6}{k}\right)$$

In their relation, f is the friction factor, Re is the Reynolds number and k is a constant determined by the concentration of the suspension. For a suspension of 0.08% concentration, $k = 0.28$. What is the value of f if Re = 3750?

Problem 2.1

f(x) = x^3 + 2x - 1

```
% Housekeeping
clear
clc

% Define initial value of x1 and x2
x1=0;
x2=1;

% Code based on the Pseudo-Code for Bisection method
while abs((x1-x2)/2)>0.0001
    x3=(x1+x2)/2;
    if (x3^3+2*x3-1)*(x1^3+2*x1-1)<0
        x2=x3;
    else
        x1=x3;
    end
end

% Output
disp((x1+x2)/2)
```

```
0.4534
```

Problem 2.2

f(x) = x^3 + 2x - 1

```
% Housekeeping
clear
clc

% Define initial value
x=1;

% Code based on the Pseudo-Code for Newton's method
while abs(x^3+2*x-1)>0.0001
    x=x-(x^3+2*x-1)/(3*x^2+2);
end

% Output
disp(x)
```

```
0.4534
```

Problem 2.3

f(x) = x^4 - 2x - 1

a) the Bisection method

```
% Housekeeping
clear
clc

% Define initial value of x1 and x2
x1=1;
x2=2;

% Code based on the Pseudo-Code for Bisection method
while abs((x1-x2)/2)>0.0001
    x3=(x1+x2)/2;
    if (x3^4-2*x3-1)*(x1^4-2*x1-1)<0
        x2=x3;
    else
        x1=x3;
    end
end

% Output
disp((x1+x2)/2)
```

```
    1.3953
```

b) Newton’s method

```
% Housekeeping
clear
clc

% Define initial value
x=1;

% Code based on the Pseudo-Code for Newton's method
while abs(x^4-2*x-1)>0.0001
    x=x-(x^4-2*x-1)/(4*x^3-2);
end

% Output
disp(x)
```

```
    1.3953
```

c) the Secant method

```
% Housekeeping
clear
clc

% Define initial value
```

```
x=1;

% Code based on the Secant method
while abs(x^4-2*x-1)>0.0001
    x1=x-(x^4-2*x-1)/(4*x^3-2);
    x2=x1-((x1^4-2*x1-1)*(x1-x))/((x1^4-2*x1-1)-(x^4-2*x-1));
    x=x2;
end

% Output
disp(x)
```

```
1.3953
```

f(x) = 2 - sin(2x) - e^x

a) the Bisection method

```
% Housekeeping
clear
clc

% Define initial value of x1 and x2
x1=0;
x2=1;

% Code based on the Pseudo-Code for Bisection method
while abs((x1-x2)/2)>0.0001
    x3=(x1+x2)/2;
    if (2-sin(2*x3)-exp(x3))*(2-sin(2*x1)-exp(x1))<0
        x2=x3;
    else
        x1=x3;
    end
end

% Output
disp((x1+x2)/2)
```

```
0.3287
```

b) Newton’s method

```
% Housekeeping
clear
clc

% Define initial value
x=2;

% Code based on the Pseudo-Code for Newton's method
```

```
while abs(2-sin(2*x)-exp(x))>0.0001
    x=x-(2-sin(2*x)-exp(x))/(-2*cos(2*x)-exp(x));
end

% Output
disp(x)
```

```
   0.3286
```

c) the Secant method

```
% Housekeeping
clear
clc

% Define initial value
x=2;

% Code based on the Secant method
while abs(x^4-2*x-1)>0.0001
    x1=x-(2-sin(2*x)-exp(x))/(-2*cos(2*x)-exp(x));
    x2=x1-((2-sin(2*x1)-exp(x1))*(x1-x))/((2-sin(2*x1)-exp(x1))-(2-sin(2*x)-exp(x)));
    x=x2;
end

% Output
disp(x)
```

```
   NaN
```

Problem 2.4

1/√ = (1/) log √ + (14 − 5.6/)

```
% Housekeeping
clear
clc

% Define initial value
x1=0;
x2=1;
k=0.28;
Re=3750;

% Code based on the Bisection method
while abs((x1-x2)/2)>(0.0008/100)
    x3=(x1+x2)/2;
    if ((1/k)*log(Re*sqrt(x3))+(14-5.6/k)-1/sqrt(x3))*((1/k)*log(Re*sqrt(x1))+(14-5.6/k)-1/sqrt
        x2=x3;
    else
        x1=x3;
```

```
    end
end

% Output
f=(x1+x2)/2;
disp(f)
```

```
0.0051
```